

Design Principles for Autoencoder Bottleneck Dimensions

Based on Intrinsic Dimension Estimation

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Abstract

Selecting the bottleneck dimension m of an autoencoder (AE) has traditionally relied on expensive manual sweeps or empirical heuristics, leaving the design rationale unclear. This paper proposes a two-stage pipeline that uses intrinsic dimension (ID) estimation as a principled guide for bottleneck design: (1) estimate the ID via TwoNN/MLE to narrow the search range, and (2) identify the optimal bottleneck dimension m^* using the AU pattern of the VAE (saturation-type: m^{sat} ; gradual-type: m^{knee}) as the primary criterion, with CKA and MSE elbow as auxiliary criteria. Validation on synthetic manifolds (Swiss Roll, Torus, Möbius Band) and MNIST yields the following **three main findings**:

1. **(RQ1) $\hat{d}_{\text{ID}}$ as a strong prior indicator:** The ID estimate $\hat{d}_{\text{ID}}$ aligns well with the optimal bottleneck dimension, particularly on synthetic manifolds, functioning as a strong prior indicator that substantially reduces trial-and-error (note: accuracy may be limited on real data).
2. **(RQ2) Correspondence between AU_{sat} and topological minimum embedding dimension:** For topologically nontrivial manifolds (Torus, Möbius Band), the VAE AU saturation value tends to correspond to the topological minimum embedding dimension (under standardized conditions; preprocessing-dependent exceptions exist).
3. **(RQ3) Isometry achieved by IsometricAE:** Decoder Jacobian regularization achieves condition number $\kappa \approx 1$, demonstrating that the prerequisite for reliable ID estimation in latent space (isometry) is practically achievable.

As a secondary finding, we quantify the temporal separation of global and local structure formation during MNIST training (Trustworthiness ≈ 0.99 by Epoch 5, followed by gradual increase in TwoNN ID). This paper distinguishes between the **local intrinsic dimension** d_{ID} (tangent-space dimension) and the **topological embedding dimension** d_{emb} ($d_{\text{emb}} \geq d_{\text{ID}}$, minimum dimension for self-intersection-free embedding).

1 Introduction

1.1 Research Background

The manifold hypothesis [1, 2] posits that the natural distribution in a high-dimensional data space $\mathbb{R}^N$ concentrates near a smooth low-dimensional manifold $\mathcal{M}$ ($\dim \mathcal{M} = d \ll N$), forming the theoretical foundation for dimensionality reduction and representation learning in machine learning. AEs are implicitly designed to capture this manifold in latent space, but the bottleneck dimension m is typically chosen by manual validation sweeps. This trial-and-error process is computationally expensive and lacks principled justification.

1.2 Research Aim

This paper proposes the principled approach of “estimating the intrinsic dimension d_{ID} of the data in advance and using it as a design guide for the bottleneck dimension.” We adopt TwoNN [3] and MLE [4] as ID estimators and clarify the quantitative relationship between their estimates and the optimal AE bottleneck dimension.

Theoretical motivation: The intuitive mechanism behind this correspondence is as follows:

- **Why AEs need at least d_{ID} dimensions:** When data lie locally on a d_{ID} -dimensional manifold, the bottleneck must satisfy $m \geq d_{\text{ID}}$ to bound reconstruction error under standard smoothness assumptions and sufficient AE capacity (see Theorem 1). Thus d_{ID} serves as a principled *lower-bound guideline* estimable before training via TwoNN/MLE.
- **Why VAE AU converges near d_{emb} :** The support $\mathbb{R}^m$ of the Gaussian prior in a VAE is topologically contractible, so embedding a manifold with nontrivial fundamental group π_1 (e.g., Torus with $\pi_1 = \mathbb{Z}^2$) without self-intersection requires $d_{\text{emb}} > d_{\text{ID}}$ dimensions [5]. Under KL regularization, AU tends to converge near d_{emb} in sufficiently capable VAEs (see §2.2, Proposition 3; with dependence on preprocessing and capacity).

In summary, the central claim of this paper is: “AEs approximately reflect d_{ID} (tangent-space dimension) while VAE AU_{sat} (AU saturation value) approximately reflects d_{emb} (topological embedding dimension) through different mechanisms.”

1.3 Two Notions of Dimension

Two fundamentally different concepts of manifold dimension appear throughout this paper:

Local Intrinsic Dimension d_{ID} The dimension of the tangent space at each point of manifold $\mathcal{M}$; a local geometric quantity measuring the degrees of freedom in a neighborhood. Both the Swiss Roll and Torus have two-dimensional tangent planes, so both have $d_{\text{ID}} = 2$. TwoNN and MLE estimate d_{ID} from local distance relations.

Topological Embedding Dimension d_{emb} The minimum dimension of Euclidean space $\mathbb{R}^{d_{\text{emb}}}$ into which $\mathcal{M}$ can be continuously embedded without self-intersection; determined by the global topological structure of the manifold (fundamental group π_1 ,

etc.) [5, 6]. For simply connected manifolds ($\pi_1 = \{1\}$, e.g., Swiss Roll): $d_{\text{emb}} = d_{\text{ID}}$; for manifolds with nontrivial π_1 (e.g., Torus: $\pi_1 = \mathbb{Z}^2$): $d_{\text{emb}} > d_{\text{ID}}$ (for Torus: $d_{\text{ID}} = 2$, $d_{\text{emb}} = 3$).

Central claim: Since TwoNN and MLE measure local distance relations, they faithfully estimate d_{ID} . On the other hand, VAE AU_{sat} , under the **continuous-mapping constraint** of neural networks, tends to approximately reflect d_{emb} [5, 7] (conditions: (i) Gaussian prior, (ii) continuous encoder/decoder, (iii) sufficient capacity, (iv) standardized preprocessing; exceptions in §6.5). This logical structure allows the two findings—“ $m^* \approx \hat{d}_{\text{ID}}$ (bottleneck design guideline, RQ1)” and “ AU_{sat} tends to correspond to d_{emb} (RQ2)” —to coexist without logical contradiction.

1.4 Research Questions

RQ1 Can TwoNN and MLE accurately estimate d_{true} on synthetic manifolds? Does the estimate align with the optimal AE bottleneck dimension m^* to a degree useful as a design guideline?

RQ2 To what extent does VAE AU_{sat} reflect the topological properties (fundamental group π_1) of the manifold?

RQ3 Can isometric embedding be achieved by decoder Jacobian regularization? How does this differ from encoder-side regularization (CAE)?

The correspondence between each RQ and experiments is summarized in Table 7 in §5.

1.5 Main Contributions

1. **Systematic evaluation of ID estimators (RQ1):** We benchmark TwoNN and MLE on synthetic manifolds and demonstrate that $\hat{d}_{\text{ID}}$ functions as a strong prior indicator, aligning well with the optimal dimension especially on synthetic manifolds (Experiments E1–E3; limitations for real data in §6.5).
2. **Observation of VAE AU_{sat} on topologically nontrivial manifolds (RQ2):** On Torus and Möbius Band, the VAE [8] AU_{sat} tends to correspond to the topological minimum embedding dimension (Experiments EA, EB; preprocessing-dependent).
3. **Isometric embedding improvement via IsometricAE (RQ3):** IsometricAE [9, 10] with decoder Jacobian regularization achieves $\kappa \approx 1$ (Experiment EC).
4. **Temporal separation of structure formation in MNIST (secondary finding):** We quantify the temporal separation in MNIST training dynamics where global neighborhood structure (Trustworthiness) forms rapidly in early training, while local ID increases gradually (Experiment E6).

1.6 Paper Structure

§2 organizes the limitations of TwoNN/MLE. §3 presents the two-stage pipeline and design rules (Fig. 1). §4 lists the main theorems and propositions. §5 covers: (5.2)

synthetic manifold experiments, (5.3) information bottleneck and topology experiments, (5.4) MNIST applications. §6 provides answers to RQ1–RQ3 and practical guidelines. §7 summarizes.

2 Related Work

2.1 Intrinsic Dimension Estimation

TwoNN [3] estimates intrinsic dimension from the ratio $\mu_i = r_2/r_1$ of the two nearest-neighbor distances. MLE [4] uses k -nearest-neighbor distances for stable estimates. DANC_o [11] uses joint distributions of angles and distances at higher computational cost. Pope et al. [12] showed natural images have far lower ID than pixel dimension. Recent work analyzed ID of deep network representations: Ansuini et al. [13] quantified layer-wise ID; Birdal et al. [14] studied ID, persistent homology, and generalization; Valeriani et al. [15] reported a “hump” structure of layer-wise ID in large transformer models.

TwoNN has a known **underestimation bias** near manifold boundaries and in high-curvature regions [3].

2.2 Limitations of TwoNN/MLE and the Need for AE-Based Approaches

TwoNN and MLE may lose accuracy due to: (1) violation of the local uniform sampling assumption in the presence of large noise or high curvature; (2) difficulty of direct application to high-dimensional input spaces ($N = 784$ for MNIST). A two-stage approach using AE latent representations is effective (see §3).

Relationship between VAE Gaussian prior and topological embedding dimension: The support $\mathbb{R}^m$ of the standard Gaussian prior is topologically **contractible** [5]. Therefore, embedding a manifold with nontrivial π_1 (e.g., Torus T^2 with $\pi_1 = \mathbb{Z}^2$) without self-intersection requires at least $d_{\text{emb}} > d_{\text{ID}}$ dimensions [5, 6]. KL regularization thus tends to drive AU_{sat} toward d_{emb} in sufficiently capable VAEs (Proposition 3, Experiments EA, EB; with dependence on capacity and preprocessing).

Differential-geometric relationship between isometry (κ) and Trustworthiness: The condition number $\kappa = \sigma_1/\sigma_m$ is a differential-geometric metric for local distance uniformity. When $\kappa = 1$, the pullback metric $G = J_g^\top J_g = I_m$, making infinitesimal distances in latent space equivalent to those in the original space (Theorem 2). $\kappa \approx 1$ is the differential-geometric refinement of Trustworthiness: it guarantees local distance preservation, not merely ordinal rank preservation.

2.3 AE Bottleneck Dimension and Representation Quality

β -VAE [16] promotes disentangled representations. Lucas et al. [17] showed AU saturates near d_{ID} in linear VAEs, but did not systematically examine topologically nontrivial cases ($\text{AU}_{\text{sat}} \approx d_{\text{emb}}$). Nazari et al. [10] and Kato et al. [9] proposed IsometricAE via decoder Jacobian regularization. The loss function is

$$\mathcal{L}_{\text{iso}} = \mathcal{L}_{\text{rec}} + \lambda_{\text{iso}} \cdot \frac{1}{n} \sum_{i=1}^n \|J_g(z_i)^\top J_g(z_i) - I_m\|_F^2 \quad (1)$$

with $\lambda_{\text{iso}} = 0.01$ in our experiments. Contractive AE (CAE) [18] penalizes the encoder Frobenius norm. Denoising AE (DAE) [19] implicitly estimates the score function.

Distinction from prior work: This paper constructs a multi-metric pipeline (ID estimation, AU, CKA, condition number κ , Trustworthiness) and demonstrates $m^* \approx \hat{d}_{\text{ID}}$ on both synthetic and real data, with explicit consideration of TwoNN’s underestimation bias.

3 Proposed Framework

We denote the bottleneck dimension consistently as m ($:= d_z$), the optimal bottleneck as m^* , and the AU saturation dimension as m^{sat} .

Key Notation Summary

Table 1: Key notation used in this paper

Symbol	Meaning	Related Section
m ($= d_z$)	AE bottleneck dimension (design parameter)	§3
m^* ($= m^*$)	Optimal bottleneck dimension	§3
m^{sat} ($= m^{\text{sat}}$)	Dimension at which AU first saturates (saturation-type)	Eq. (4)
m^{knee} ($= m^{\text{knee}}$)	AU growth inflection point (gradual-type)	Eq. (5)
AU_{sat}	Maximum active units reached on grid	Eq. (4)
$\text{AU}(m)$	Active unit count at bottleneck m	§3
$\hat{d}_{\text{ID}}$	ID estimate from TwoNN/MLE	§3
d_{true}	True ID of synthetic manifold (known)	§5.2
d_{emb}	Topological minimum embedding dimension ($d_{\text{emb}} \geq d_{\text{ID}}$)	§1.3
κ	Condition number of decoder Jacobian (isometry metric)	§2.2
$r(m)$	Normalized AU growth rate	Eq. (2)
τ_{knee}	Inflection-point threshold (0.25 in this work)	Eq. (5)

3.1 Two-Stage Pipeline

Stage 1: ID Estimator Evaluation and Selection. Benchmark TwoNN and MLE on synthetic manifolds with known d_{true} to select a reliable estimator. For real data with large input dimension N , apply TwoNN/MLE to the latent space of a reference AE trained with a large bottleneck ($m_{\text{ref}} \gg \hat{d}_{\text{ID}}$).

Stage 2: Optimal Bottleneck Search via VAE Sweep. Sweep m over a discrete grid $\mathcal{M} = \{1, 2, 4, 8, 12, 16, 20, 32, 64, 128, 256\}$, training a VAE ($\beta = 4$) at each m and computing $\text{AU}(m)$.

Remark 1 (Grid Resolution and Accuracy of m^*). *Since $\mathcal{M}$ is an approximately log-scale grid, when d_{ID} falls between grid points (e.g., $d_{\text{ID}} = 6$ between $m = 4$ and $m = 8$), the grid point where AU first exceeds d_{ID} serves as a conservative upper bound for m^* . Practical recommendation: verify with a finer grid near $\hat{d}_{\text{ID}}$.*

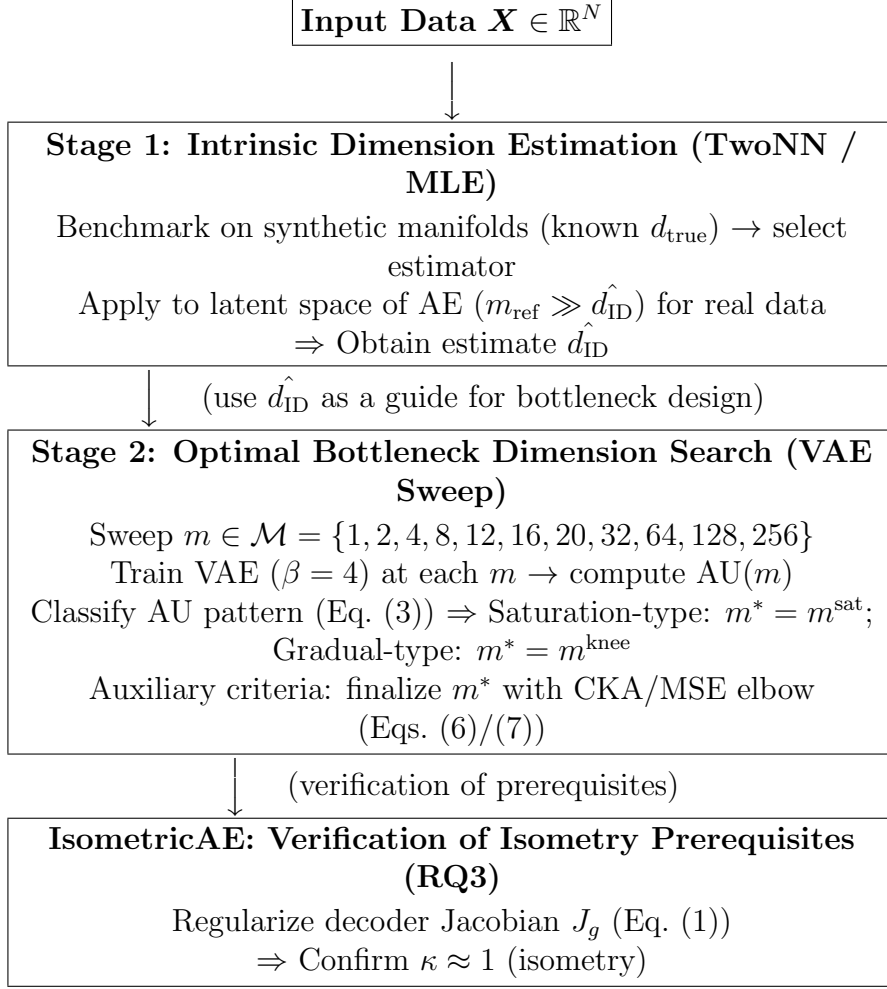


Figure 1: Data flow of the proposed framework. Stage 1 estimates $\hat{d}_{\text{ID}}$, which guides the VAE bottleneck search in Stage 2. IsometricAE (RQ3) verifies that the isometry condition for reliable latent-space ID estimation is achievable.

Remark 2 (Circularity and Isometry Bias in ID Estimation for Real Data). *For large- N data (e.g., MNIST with $N = 784$), we apply TwoNN/MLE to the latent space of a reference AE ($m = 256$) rather than directly to the input. Since this reference AE lacks isometric regularization, its latent space may have $\kappa > 1$, introducing potential bias in $\hat{d}_{\text{ID}}$. Empirical mitigation: convergence of TwoNN ID (9.54–10.02) and MLE ID (11.26–11.85) across multiple $m \geq 64$ values (Table 17) suggests the bias is not systematically unidirectional. This procedure is a one-time reference step; the final model is designed separately with $m \approx \hat{d}_{\text{ID}}$.*

3.2 Optimal Bottleneck Dimension

The normalized AU growth rate is defined as

$$r(m) = \frac{\text{AU}(m) - \text{AU}(m_{\text{prev}})}{m - m_{\text{prev}}} \quad (2)$$

where m_{prev} is the preceding grid point in $\mathcal{M}$.

AU pattern classification:

- **Saturation-type:** AU maintains the same value at two or more consecutive grid points (plateau width ≥ 2). Typical examples: Swiss Roll, Torus.
- **Gradual-type:** AU increases nearly monotonically; $\text{AU}(m_{\max}) \approx \text{AU}_{\text{sat}}$ only at the largest m . Typical example: MNIST.

Case-based definition of m^* :

$$m^* = \begin{cases} m^{\text{sat}} := \min\{m \in \mathcal{M} \mid \text{AU}(m) = \text{AU}_{\text{sat}}\} & (\text{saturation-type}) \\ m^{\text{knee}} := \min\{m \in \mathcal{M} \mid r(m) < \tau_{\text{knee}} \cdot r_{\max}\} & (\text{gradual-type}) \end{cases} \quad (3)$$

where

$$\text{AU}_{\text{sat}} = \max_{m' \in \mathcal{M}} \text{AU}(m'), \quad (4)$$

$$r_{\max} = \max_{m' \in \mathcal{M}} r(m'), \quad \tau_{\text{knee}} = 0.25. \quad (5)$$

Design Rule (two-branch):

Saturation-type: Given the saturation set $\mathcal{C}_{\text{sat}} = \{m \in \mathcal{M} \mid \text{AU}(m) = \text{AU}_{\text{sat}}\}$,

$$m^* = \min \left\{ m \in \mathcal{C}_{\text{sat}} \mid \text{CKA}(m) \geq (1 - \varepsilon) \max_{m' \in \mathcal{C}_{\text{sat}}} \text{CKA}(m') \right\} \quad (6)$$

with $\varepsilon = 0.01$. Verify consistency with the MSE elbow m_{elbow}^* (Eq. (9)).

Gradual-type: Given $\mathcal{C}_{\text{knee}} = \{m \in \mathcal{M} \mid m \geq m^{\text{knee}}, r(m) < \tau_{\text{knee}} \cdot r_{\max}\}$,

$$m^* = \min \left\{ m \in \mathcal{C}_{\text{knee}} \mid \text{CKA}(m) \geq (1 - \varepsilon) \max_{m' \in \mathcal{C}_{\text{knee}}} \text{CKA}(m') \right\} \quad (7)$$

Supplementary MSE Elbow (with smoothing): After moving-average smoothing (window $w = 1$): $\tilde{\mathcal{L}}(m_k) = \frac{1}{2w+1} \sum_{j=-w}^w \mathcal{L}(m_{k+j})$, the normalized second difference is

$$\Delta^2(m_k) = \frac{\tilde{\mathcal{L}}(m_{k-1}) - 2\tilde{\mathcal{L}}(m_k) + \tilde{\mathcal{L}}(m_{k+1}))}{\tilde{\mathcal{L}}(m_1)}, \quad (8)$$

$$m_{\text{elbow}}^* = \arg \max_{k \in \{2, \dots, K-1\}} \Delta^2(m_k). \quad (9)$$

3.3 Evaluation Metrics

TwoNN ID estimator [3]: $\hat{d}_{\text{ID}}^{\text{TwoNN}} = 1/\mathbb{E}[\log \mu_i]$, where $\mu_i = r_2/r_1$.

MLE ID estimator [4]: $\hat{d}_{\text{ID}}^{\text{MLE}} = \left[\frac{1}{k-1} \sum_{j=1}^{k-1} \log \frac{r_k(\mathbf{x}_i)}{r_j(\mathbf{x}_i)} \right]^{-1}$.

Active Units (AU) [17]:

$$\text{AU} = \sum_{k=1}^m \mathbf{1}[V_k > \delta], \quad \delta = 10^{-2}, \quad V_k = \text{Var}_{\mathbf{x}}[\mathbb{E}_{q_{\phi}}[z_k | \mathbf{x}]]. \quad (10)$$

CKA [20]: With $K = \mathbf{X}\mathbf{X}^{\top}$, $L = \hat{\mathbf{X}}\hat{\mathbf{X}}^{\top}$, $H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^{\top}$:

$$\text{CKA}(K, L) = \frac{\text{HSIC}(K, L)}{\sqrt{\text{HSIC}(K, K) \cdot \text{HSIC}(L, L)}}, \quad (11)$$

where $\text{HSIC}(K, L) = \frac{1}{(n-1)^2} \text{tr}(KHLH)$.

Condition number κ : $\kappa = \sigma_1/\sigma_m$ (largest/smallest singular values of $J_g(z) \in \mathbb{R}^{N \times m}$).

Trustworthiness: Fraction of latent-space neighborhoods that correspond to neighborhoods in the original space [12].

3.4 Theoretical Framework for Multi-Class Data: Manifold Entanglement

For data like MNIST with multiple classes forming a composite structure, we introduce the concept of **Manifold Entanglement** [7]: the data support is $\mathcal{X} = \bigcup_{k=0}^{K-1} \mathcal{M}_k$ where $\{\mathcal{M}_k\}$ are mutually disconnected smooth sub-manifolds. This framework predicts for MNIST: (1) amplified TwoNN underestimation bias; (2) $\text{AU}_{\text{sat}} > d_{\text{ID}}$ due to class-boundary representation; (3) smooth MSE decrease (smooth elbow problem).

4 Theoretical Background

The following theorems and propositions provide the theoretical interpretation of our experiments. **Theorems** (Theorem 1 and Theorem 2) are given with proofs. **Propositions** (Propositions 1–4) are theoretical interpretations emphasizing intuitive meaning (rigorous general proofs are omitted).

Theorem 1 (Bottleneck Lower Bound). *For a data distribution $p(\mathbf{x})$ concentrated on a smooth d -dimensional manifold $\mathcal{M} \subset \mathbb{R}^N$, having bounded expected reconstruction error requires the AE bottleneck dimension m to satisfy $m \geq d$.*

Proof. Consider an AE with encoder $f : \mathbb{R}^N \rightarrow \mathbb{R}^m$ and decoder $g : \mathbb{R}^m \rightarrow \mathbb{R}^N$. Assume $m < d$. Since f is continuous, the image of $g \circ f : \mathbb{R}^N \rightarrow \mathbb{R}^N$ is contained in $g(\mathbb{R}^m)$, a set of dimension at most m . Since the smooth d -dimensional manifold $\mathcal{M}$ has $d > m$ -dimensional volume, the intersection of any set of dimension $\leq m$ with $\mathcal{M}$ is a null set with respect to the Lebesgue measure on $\mathcal{M}$ (a consequence of Sard’s theorem). Therefore, under the assumption that $p(\mathbf{x})$ is concentrated on $\mathcal{M}$, $g(f(\mathbf{x})) = \mathbf{x}$ cannot hold p -almost everywhere, so $\mathbb{E}_p[\|\mathbf{x} - g(f(\mathbf{x}))\|^2] > 0$ cannot be bounded to zero. Hence $m \geq d$ is necessary. $\square$

Experiment E1 (Swiss Roll, $d_{\text{true}} = 2$) confirms this: MSE improves approximately 35-fold ($1.016 \rightarrow 0.029$, 3-seed average) at the $m = 1 \rightarrow 2$ transition (§5.2).

Proposition 1 (Effective Rank of Decoder Jacobian). *When a sufficiently capable AE achieves accurate reconstruction of $\mathcal{M}$, the number of significant singular values of $J_g(z) \in \mathbb{R}^{N \times m}$ equals $r^* = \min(m, d_{\text{emb}})$. For $m \leq d_{\text{emb}}$: $\text{rank } J_g(z) = m$; for $m > d_{\text{emb}}$: surplus dimensions degenerate and $\text{rank } J_g(z) = d_{\text{emb}}$.*

Experiment E2 confirms: for Torus at $m = 4 > d_{\text{emb}} = 3$, the significant singular value count is 3.00.

Theorem 2 (Pullback Metric and Isometry). *When decoder $g : \mathbb{R}^m \rightarrow \mathbb{R}^N$ is an isometric embedding into Riemannian manifold $(\mathcal{M}, g_{\mathcal{M}})$, the pullback metric $G(z) = J_g(z)^\top J_g(z)$ is proportional to the identity, i.e., $\kappa = 1$.*

Proof. By definition of an isometric embedding, for any $z \in \mathbb{R}^m$ and tangent vectors $u, v \in \mathbb{R}^m$: $(g_{\mathcal{M}})_{g(z)}(J_g(z)u, J_g(z)v) = \langle u, v \rangle = u^\top v$. From the definition of the pullback metric $(g^*g_{\mathcal{M}})_z(u, v) := (g_{\mathcal{M}})_{g(z)}(J_g(z)u, J_g(z)v) = u^\top G(z)v$, we have $u^\top G(z)v = u^\top v$ for all u, v , hence $G(z) = I_m$, i.e., $\kappa = \sigma_{\text{max}}/\sigma_{\text{min}} = 1$. $\square$

Experiment EC shows IsometricAE achieves $\kappa \approx 1.055 \pm 0.005$.

Proposition 2 (Noise Selectivity). *The optimal denoising AE approximates the score function $\nabla_{\mathbf{x}} \log p(\mathbf{x})$. An AE trained with $m = d_{\text{true}}$ selectively removes noise in the normal direction more than in the tangent direction.*

Experiment E3 confirms: at $m = d_{\text{true}}$, normal-direction error $>$ tangent-direction error.

Proposition 3 (KL Regularization Required for AU Saturation). *For a Vanilla AE (no regularization), $AU = m$ for any $m > d_{\text{ID}}$. AU saturation requires KL regularization (the information bottleneck constraint).*

Experiment EA confirms: only VAE ($\beta = 4$) exhibits AU saturation.

Proposition 4 (Neighborhood Preservation and Optimal Bottleneck). *At $m = d_{\text{true}}$ (or the topological minimum embedding dimension), Trustworthiness improves sharply, and further increases in m yield limited additional improvement.*

Experiment E5 confirms: Swiss Roll at $m = 2$ and Torus at $m = 3 = d_{\text{emb}}$, Trustworthiness jumps ($\Delta \approx 0.015$), with < 0.001 improvement beyond.

Table 2 summarizes the theorem/proposition–experiment correspondences.

Table 2: Theorem/Proposition–Experiment Correspondence

Theorem/Proposition	Claim	Exper
Theorem 1 (bottleneck lower bound)	MSE spikes for $m < d_{\text{true}}$	E1
Proposition 1 (effective rank)	Significant SVs = $\min(m, d_{\text{emb}})$	E2
Theorem 2 (pullback metric)	$\kappa \approx 1$ isometry	EC
Proposition 2 (noise selectivity)	Normal direction selectively removed	E3
Proposition 3 (AU saturation)	KL regularization required	EA
Proposition 4 (neighborhood preservation)	Trustworthiness jumps at $m = d_{\text{true}}$ (or d_{emb})	E5
Topological min. embedding (supplementary)	$AU_{\text{sat}} \approx d_{\text{emb}}$ (conditional)	EB

5 Experiments

5.1 Experimental Setup

Common hyperparameters are shown in Table 3; model-specific hyperparameters in Tables 4 and 5.

Table 3: Common Hyperparameters

Parameter	Value
Optimizer	Adam ($\text{lr} = 1 \times 10^{-3}$)
Batch size	128
Activation	ReLU (output layer: Sigmoid)
Random seed	42 (fixed)
Framework	PyTorch 2.0

Table 4: Synthetic Manifold AE/VAE Hyperparameters

Parameter	Value
Hidden dims	[64, 32]
Training epochs	200–300
VAE β	4.0
IsometricAE λ_{iso}	0.01
Sample size	5,000 (train 4,000 + test 1,000)

Table 5: MNIST AE/VAE Hyperparameters

Parameter	Value
Input dim	784 (28×28 flattened)
Hidden dims	[512, 256]
Training epochs	300 (E4), 200 (E6)
VAE β	4.0
Bottleneck m	1, 2, 4, 8, 12, 16, 20, 32, 64, 128, 256
Batch size	128

ID estimation uses $k = 2$ for TwoNN and $k = 10$ for MLE. The AU threshold $\delta = 10^{-2}$ follows [17].

AU threshold sensitivity analysis: We compared AU counts across $\delta \in \{0.001, 0.01, 0.05, 0.1\}$ for Swiss Roll and Torus VAE ($\beta = 4$). As shown in Table 6, AU saturation values are *completely robust* to the choice of δ .

Table 6: AU Threshold δ Sensitivity (VAE $\beta = 4$): AU values are completely robust

m	Swiss Roll				Torus			
	$\delta=0.001$	$\delta=0.01$	$\delta=0.05$	$\delta=0.1$	$\delta=0.001$	$\delta=0.01$	$\delta=0.05$	$\delta=0.1$
2	2	2	2	2	2	2	2	2
3	2	2	2	2	2	2	2	2
6	2	2	2	2	3	3	3	3
12	2	2	2	2	3	3	3	3
16	2	2	2	2	3	3	3	3

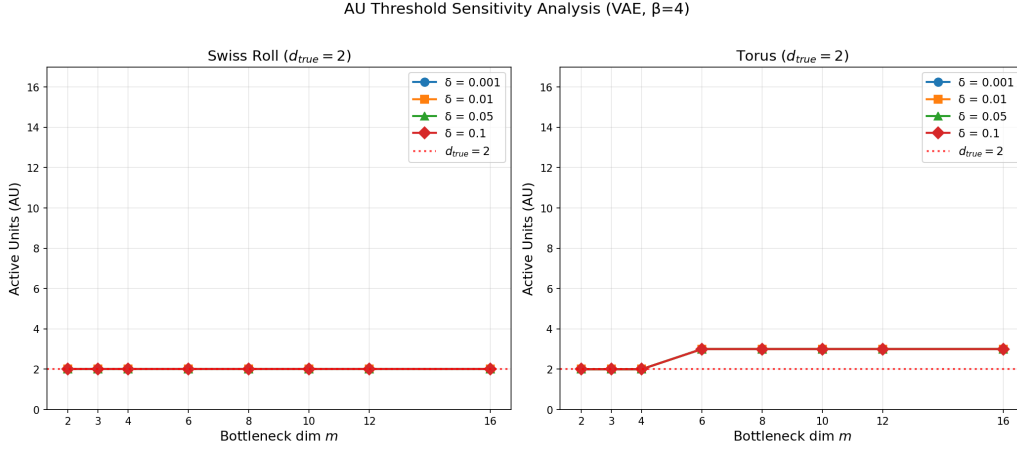


Figure 2: AU threshold δ sensitivity (VAE $\beta = 4$). AU saturation patterns are identical across all δ values for both Swiss Roll (left) and Torus (right).

Table 7 lists the experiment–RQ correspondence.

Exp.	RQ	Section	Purpose
E1	RQ1	§5.2	ID accuracy; $m^* \approx \hat{d}_{\text{ID}}$
E2	Prop. 1/Thm. 2 basis	§5.2	Decoder Jacobian rank and κ
E3	Prop. 2	§5.2	Noise selectivity
E5	Prop. 4	§5.2.4	Trustworthiness vs. m
EA	RQ2	§5.3	VAE AU saturation via KL regularization
EB	RQ2 (depth)	§5.3	AU_{sat} vs. topological embedding dim.
EC	RQ3	§5.3	Isometric embedding achievability
E4	RQ1 (real data)	§5.4	MNIST $m^* \approx \hat{d}_{\text{ID}}$
E6	Novel finding	§5.4	MNIST training dynamics
EX	Statistical reliability	§5.4	Multi-seed stability

5.2 ID Estimation and Optimal Bottleneck on Synthetic Manifolds (RQ1)

Question answered (RQ1): Does TwoNN/MLE estimate d_{true} accurately, and does $\hat{d}_{\text{ID}}$ align with m^* as a design guideline?

5.2.1 Reconstruction Error Elbow Analysis (Experiment E1)

We swept m for Swiss Roll ($d_{\text{true}} = 2$, ambient = 3) and Torus ($d_{\text{true}} = 2$, ambient = 3), training AEs and recording MSE, AU, TwoNN ID, and MLE ID (3-seed mean $\pm$ std).

Table 8: E1: Swiss Roll ($d_{\text{true}} = 2$), 3-seed mean $\pm$ std

m	MSE	AU	TwoNN ID	MLE ID
1	1.016 ± 0.315	1.0 ± 0.0	1.00 ± 0.00	1.18 ± 0.02
2 = d_{true}	0.029 ± 0.001	2.0 ± 0.0	1.15 ± 0.05	1.12 ± 0.00
3	0.001 ± 0.000	3.0 ± 0.0	1.94 ± 0.06	2.09 ± 0.02
4	0.001 ± 0.000	4.0 ± 0.0	1.95 ± 0.06	2.10 ± 0.01

Table 9: E1: Torus ($d_{\text{true}} = 2$), 3-seed mean $\pm$ std

m	MSE	AU	TwoNN ID	MLE ID
1	0.356 ± 0.022	1.0 ± 0.0	1.01 ± 0.01	1.12 ± 0.00
2	0.109 ± 0.007	2.0 ± 0.0	1.86 ± 0.04	2.11 ± 0.02
3 (eff. elbow)	0.000 ± 0.000	3.0 ± 0.0	2.02 ± 0.00	2.27 ± 0.03
4	0.000 ± 0.000	4.0 ± 0.0	2.00 ± 0.03	2.28 ± 0.02

For Swiss Roll, MSE drops approximately 34-fold ($1.016 \rightarrow 0.029$) at $m = 2 = d_{\text{true}}$, confirming Theorem 1. TwoNN ID at $m = 2$ (1.15 ± 0.05) underestimates the true ID of 2; this is the known finite-sample boundary bias [3]. At $m = 3$, TwoNN ID converges to 1.94 ± 0.06 and MLE to 2.09 ± 0.02 .

For Torus, the effective MSE elbow appears at $m = 3$ despite $d_{\text{true}} = 2$. The topological explanation is discussed in §6.2.

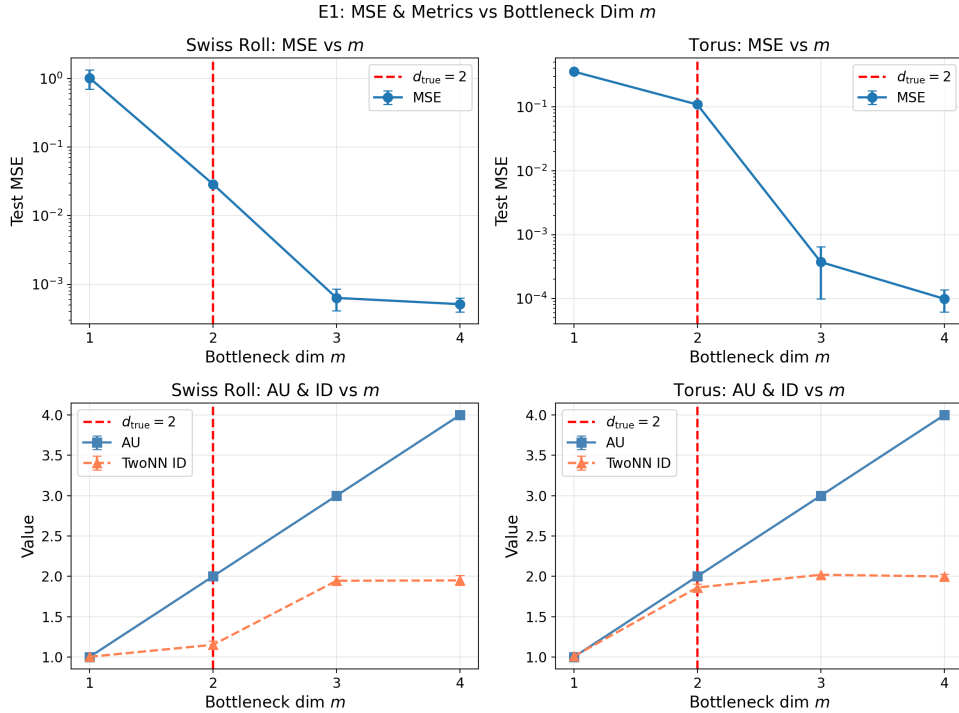


Figure 3: E1: MSE vs. m (3-seed mean $\pm$ std, log scale). Swiss Roll (solid, $\bullet$): sharp drop at $m = 2 = d_{\text{true}}$. Torus (dashed, $\blacktriangle$): effective elbow at $m = 3$.

5.2.2 Decoder Jacobian Analysis (Experiment E2)

Table 10: E2: Decoder Jacobian Analysis

Manifold	m	Sig. SVs	κ	TSA error ($^\circ$)
Swiss Roll	1	1.00	1.00	26.2
	2	2.00	1.51	44.3
	3	3.00	1.75	38.9
	4	3.00	2.03	43.6
Torus	1	1.00	1.00	10.6
	2	2.00	2.30	21.8
	3	3.00	1.50	22.0
	4	3.00	1.47	25.2

For Torus at $m = 4$, significant SVs = 3.00 $\neq$ 4, confirming Proposition 1: surplus dimension degenerates to $d_{\text{emb}} = 3$.

5.2.3 Noise Selectivity (Experiment E3)

Table 11: E3: NSR Analysis (Swiss Roll, $\sigma = 0.1$)

m	NSR mean	NSR median	Tangent err.	Normal err.
1	6.19	0.586	0.153	0.128
2 = d_{true}	1.01	0.562	0.079	0.126
3	1.48	1.010	0.121	0.127
4	1.46	0.988	0.120	0.129

At $m = 2 = d_{\text{true}}$, normal-direction error (0.126) $>$ tangent-direction error (0.079), confirming Proposition 2.

5.2.4 Trustworthiness vs. Bottleneck Dimension (Experiment E5)

Table 12: E5: Trustworthiness vs. m (3-seed mean $\pm$ std, $k = 10$)

m	Swiss Roll ($d_{\text{true}} = 2$)	Torus ($d_{\text{true}} = 2, d_{\text{emb}} = 3$)
1	0.9838 ± 0.0062	0.9395 ± 0.0043
2	0.9990 ± 0.0000 $\leftarrow$	0.9847 ± 0.0018
3	0.9995 ± 0.0001	0.9996 ± 0.0001 $\leftarrow$
4	0.9993 ± 0.0001	0.9995 ± 0.0002

Swiss Roll: Trustworthiness jumps from 0.984 $\rightarrow$ 0.999 at $m = 1 \rightarrow 2$ ($\Delta \approx 0.015$). Torus: maximum improvement at $m = 2 \rightarrow 3$ ($\Delta \approx 0.015$), consistent with $d_{\text{emb}} = 3$ (Proposition 4).

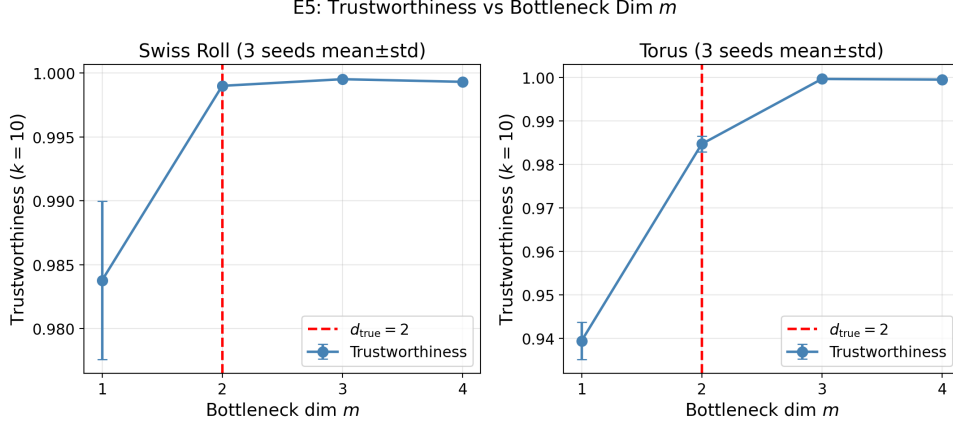


Figure 4: E5: Trustworthiness vs. m (3-seed mean $\pm$ std). Swiss Roll (left): sharp improvement at $m = 2$. Torus (right): sharp improvement at $m = 3 = d_{\text{emb}}$.

5.3 Information Bottleneck and Topological Embedding (RQ2, RQ3)

5.3.1 AU Saturation via Information Bottleneck (Experiment EA)

Table 13: EA: Swiss Roll VAE AU vs. m (3-seed mean $\pm$ std, $\beta = 4$)

m	VAE (AU)	VAE MSE
1	1.0 ± 0.0	1.492 ± 0.257
2	2.0 ± 0.0	0.070 ± 0.012
3	$2.0 \pm 0.0 \leftarrow \text{sat.}$	0.054 ± 0.004
6	2.0 ± 0.0	0.043 ± 0.006
12	2.0 ± 0.0	0.040 ± 0.003
16	2.0 ± 0.0	0.037 ± 0.003

Table 14: EA: Torus VAE AU vs. m (3-seed mean $\pm$ std, $\beta = 4$)

m	VAE (AU)	VAE MSE
1	1.0 ± 0.0	0.565 ± 0.029
2	2.0 ± 0.0	0.299 ± 0.009
4	2.0 ± 0.0	0.219 ± 0.005
6	$3.0 \pm 0.0 \leftarrow \text{sat.}$	0.074 ± 0.010
12	3.0 ± 0.0	0.016 ± 0.001
16	3.0 ± 0.0	0.010 ± 0.001

For Swiss Roll, VAE AU maintains $\text{AU} = 2$ ($= d_{\text{true}}$) for all $m \geq 2$ with zero std—strong evidence of robustness. For Torus, VAE AU saturates at $3 > d_{\text{true}} = 2$ for $m \geq 6$, corresponding to the topological minimum embedding dimension of 3. Vanilla AE always has $\text{AU} = m$, confirming Proposition 3.

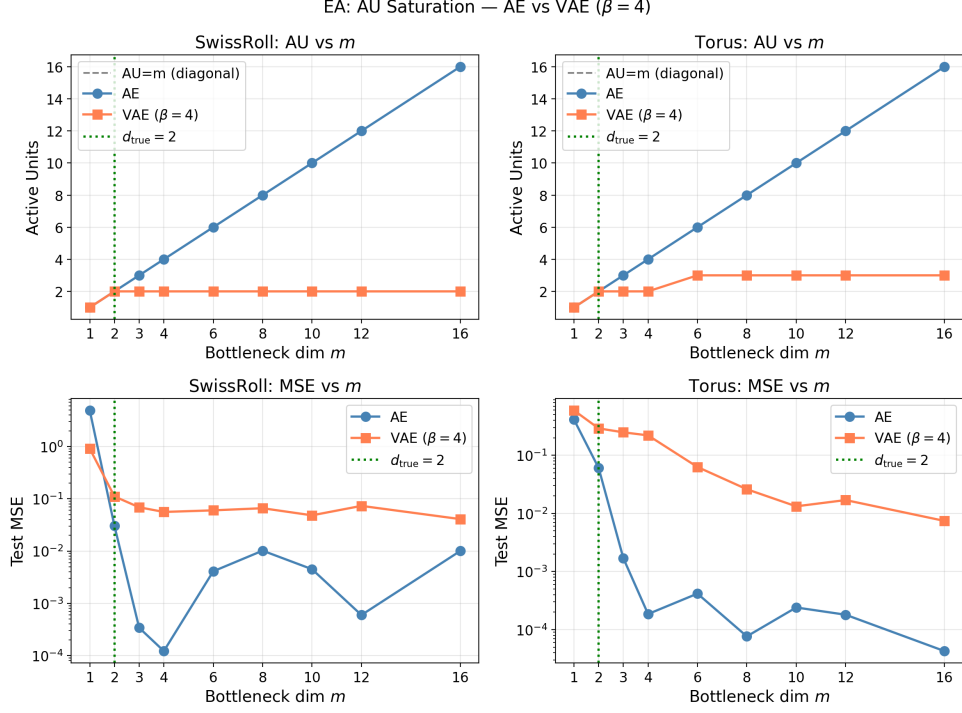


Figure 5: EA: AU vs. m for Swiss Roll (AE vs. VAE). Only VAE exhibits AU saturation at $d_{\text{true}} = 2$.

5.3.2 Topological Constraints and AU Saturation (Experiment EB)

Table 15: EB: Topological AU Experiment (VAE $\beta = 4$, standardized inputs). $\checkmark$: $\text{AU}_{\text{sat}} = \text{topological embedding dim.}$ $\times$: exception. $\triangle$: conditional.

Manifold	d_{true}	π_1	Topol. d_{emb}	VAE AU_{sat}
Torus T^2	2	$\mathbb{Z}^2$	3	3 $\checkmark$
Möbius Band	2	$\mathbb{Z}$	3	3 $\checkmark$
Swiss Roll	2	$\{1\}$	2	3 $\times$
Circle S^1	1	$\mathbb{Z}$	2	2 ($m \geq 6$) $\triangle$

Torus and Möbius Band show $\text{AU}_{\text{sat}} = d_{\text{emb}} = 3$ ($\checkmark$: main result). Swiss Roll shows $\text{AU}_{\text{sat}} = 3 \neq d_{\text{emb}} = 2$ under standardization ($\times$: preprocessing-dependent exception).

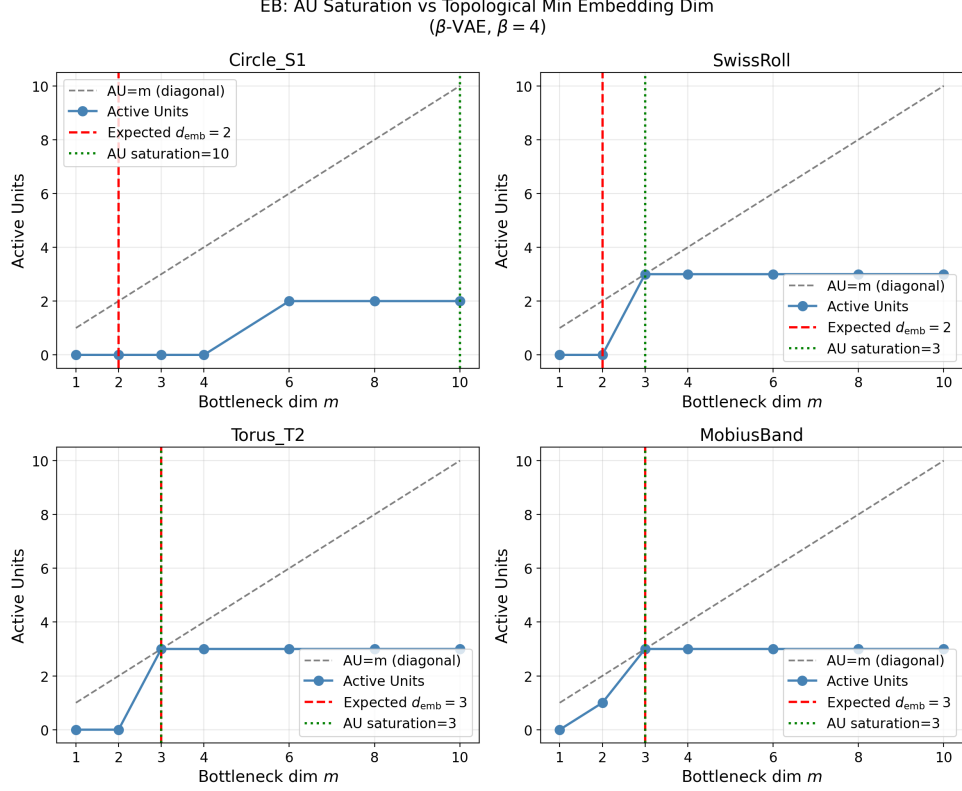


Figure 6: EB: VAE AU saturation for four manifolds (standardized inputs). Torus and Möbius Band (✓) are the main results.

5.3.3 Isometric Embedding via Decoder Regularization (Experiment EC)

Table 16: EC: IsometricAE Comparison (Swiss Roll, $m = 2, 3$ -seed mean $\pm$ std)

Model	MSE	κ	Trustworthiness
AE	0.0276 ± 0.0005	1.433 ± 0.182	0.9990 ± 0.0000
CAE	0.0289 ± 0.0011	1.907 ± 0.059	0.9990 ± 0.0000
IsometricAE	0.0283 ± 0.0005	1.055 ± 0.005	0.9990 ± 0.0000
DAE	0.0303 ± 0.0018	1.431 ± 0.144	0.9990 ± 0.0000

IsometricAE achieves the lowest condition number $\kappa = 1.055 \pm 0.005$, confirming Theorem 2. CAE shows the highest $\kappa = 1.907 \pm 0.059$: encoder-side regularization unexpectedly distorts the decoder pullback metric. All models achieve Trustworthiness = 0.9990 ± 0.0000 .

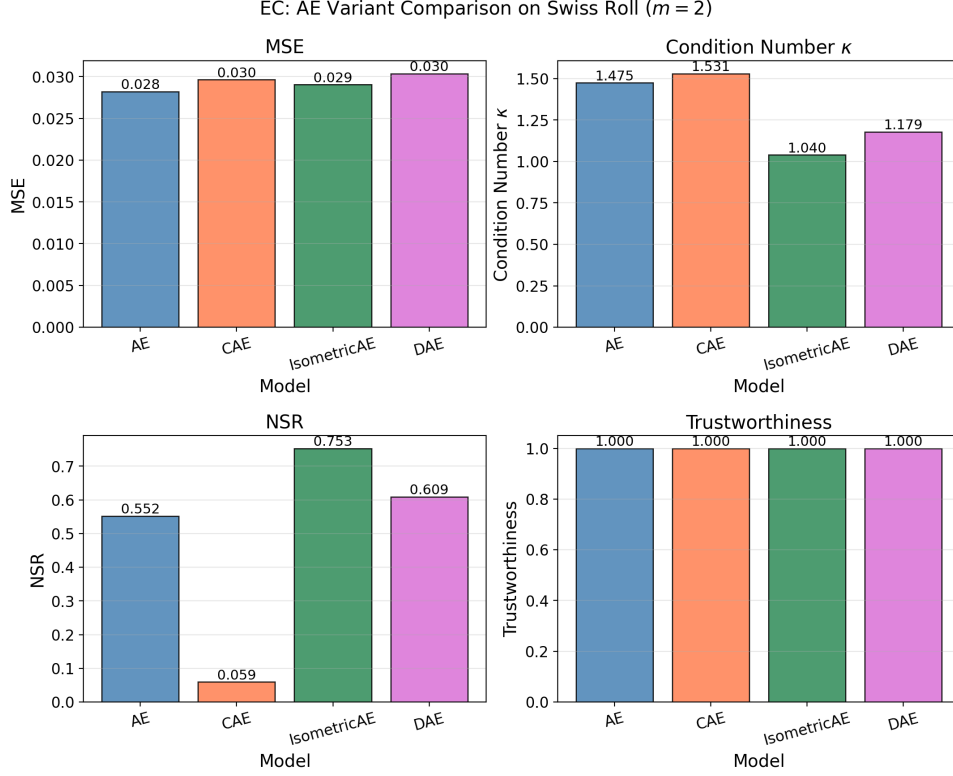


Figure 7: EC: Condition number κ comparison. IsometricAE achieves the lowest $\kappa \approx 1$.

5.4 Application to MNIST (RQ1 + Novel Finding)

MNIST (10 classes) has a complex composite manifold structure (Manifold Entanglement, §3.4).

5.4.1 MNIST Bottleneck Sweep (Experiment E4)

Table 17: E4: MNIST AE Results (300 epochs). TwoNN ID and MLE ID converge for $m \geq 64$, indicating effective ID ≈ 10 –12.

m	MSE	AU	TwoNN ID	MLE ID	CKA
1	0.0437	1	1.02	1.15	0.102
4	0.0292	4	3.86	4.29	0.575
8	0.0191	8	6.08	6.70	0.725
12	0.0148	12	7.59	7.94	0.791
16	0.0121	16	7.96	8.61	0.812
32	0.0087	32	8.84	9.89	0.875
64	0.0062	64	9.54	11.26	0.908
128	0.0061	128	9.84	11.45	0.921
256	0.0057	256	10.02	11.85	0.953

Table 18: E4: MNIST VAE ($\beta = 4$) Results (300 epochs). At $m = 12$, AU growth rate drops sharply ($m^{\text{knee}} = 12$). MNIST is gradual-type, so $m^{\text{knee}} \neq \text{AU}_{\text{sat}}$.

m	MSE	AU	TwoNN ID	MLE ID
1	0.0464	1	1.00	1.11
4	0.0275	4	3.72	4.28
8	0.0186	8	6.36	6.98
12	0.0184	8	6.26	6.97
16	0.0148	11	7.18	8.23
32	0.0121	15	8.25	9.17
64	0.0103	29	9.16	9.87
128	0.0085	26	10.02	10.72
256	0.0072	36	10.14	11.60

From Table 17, TwoNN ID (10.02) and MLE ID (11.85) converge near $m \geq 64$, suggesting effective ID $\approx \mathbf{10-12}$. At $m = 12$, AU growth rate drops sharply ($m^{\text{knee}} = 12$), consistent with $\hat{d}_{\text{ID}} \approx 10-12$ (Eq. (7)).

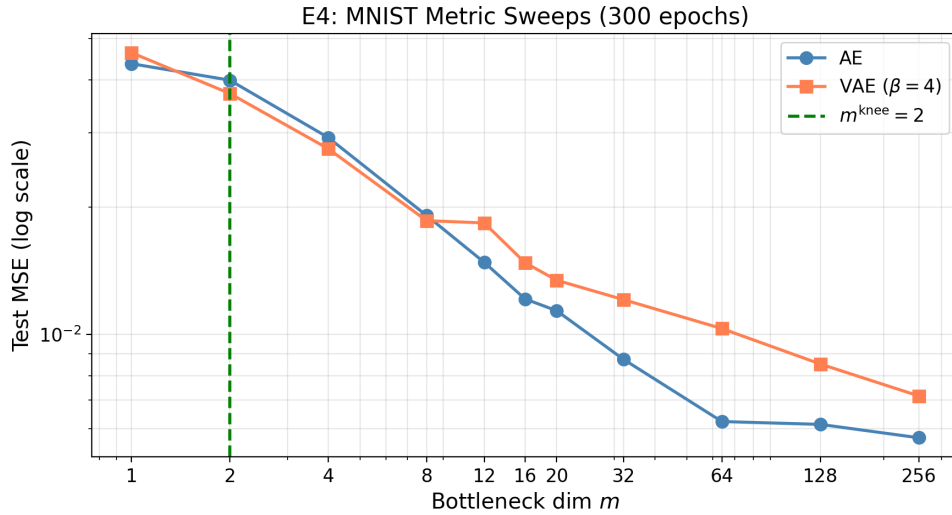


Figure 8: E4: MNIST metric sweeps (300 epochs). Arrow marks $m^{\text{knee}} = 12$. MSE shows smooth decrease (smooth elbow problem).

5.4.2 MNIST Training Dynamics (Experiment E6)

Table 19: E6: MNIST Training Dynamics ($m = 16$)

Epoch	MSE (test)	Trustworthiness	CKA	TwoNN ID
1	0.04631	0.9263	0.785	4.81
2	0.03575	0.9665	0.832	5.71
5	0.02485	0.9882	0.872	6.67
10	0.02031	0.9912	0.870	6.87
50	0.01576	0.9930	0.853	7.10
100	0.01472	0.9933	0.844	7.52
200	0.01411	0.9929	0.831	7.65

Trustworthiness improves rapidly ($0.9263 \rightarrow 0.9882$ in Epochs 1–5), while TwoNN ID increases gradually ($4.81 \rightarrow 7.65$), demonstrating temporal separation of global and local structure formation.

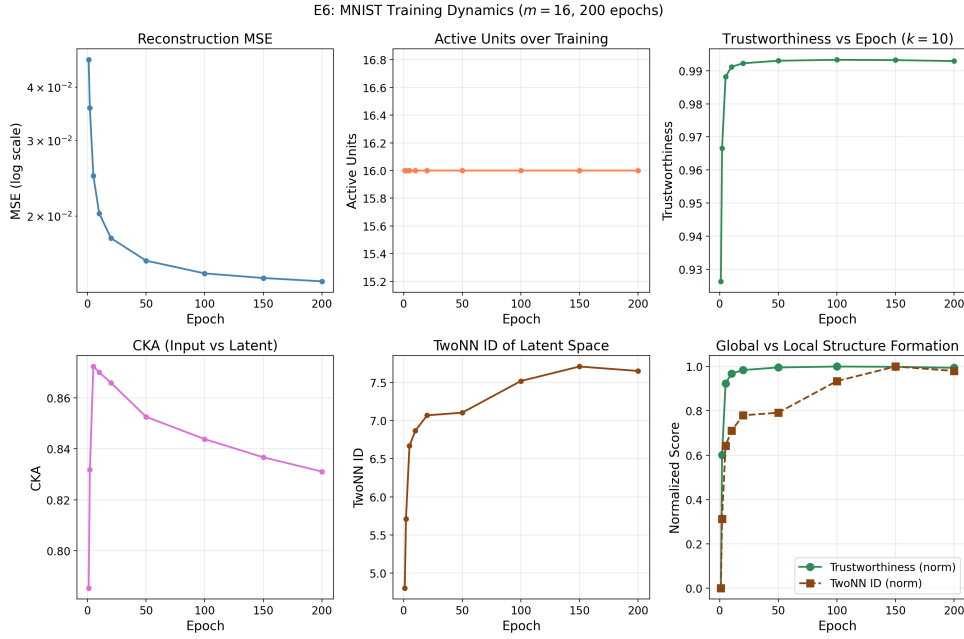


Figure 9: E6: MNIST training dynamics ($m = 16$). Trustworthiness (global) forms rapidly by Epoch 5; TwoNN ID (local) increases gradually.

5.4.3 Multi-Seed Statistical Reliability (Experiment EX)

Table 20: EX: MNIST AE Multi-Seed Stability (3-seed mean $\pm$ std, 150 epochs)

m	MSE	AU	TwoNN ID
8	0.0176 ± 0.0001	8.0 ± 0.0	6.53 ± 0.02
12	0.0132 ± 0.0002	12.0 ± 0.0	7.87 ± 0.18
16	0.0108 ± 0.0001	16.0 ± 0.0	8.58 ± 0.20
20	0.0096 ± 0.0002	20.0 ± 0.0	8.99 ± 0.07
32	0.0070 ± 0.0001	32.0 ± 0.0	9.72 ± 0.16

Table 21: EX: MNIST VAE Multi-Seed Stability (3-seed mean $\pm$ std, 150 epochs)

m	MSE	AU	TwoNN ID
8	0.0175 ± 0.0001	8.0 ± 0.0	6.40 ± 0.10
12	0.0153 ± 0.0005	9.7 ± 0.5	7.20 ± 0.11
16	0.0133 ± 0.0005	12.0 ± 0.8	7.99 ± 0.20
20	0.0123 ± 0.0005	13.3 ± 0.9	8.33 ± 0.15
32	0.0107 ± 0.0003	16.7 ± 0.9	9.18 ± 0.28

AE MSE std is within ± 0.0002 and TwoNN ID within ± 0.20 , confirming high reproducibility.

5.4.4 Quantitative Verification of Manifold Entanglement: Class-Restricted MNIST (Experiment EY)

Objective: To empirically test the Manifold Entanglement hypothesis (§3.4), we vary the number of MNIST classes $K \in \{1, 2, 5, 10\}$ and examine whether AU_{sat} and TwoNN ID increase systematically with K . $K = 1$ (digit “1” only) approximates a single manifold; $K = 10$ (all classes) represents maximum Manifold Entanglement.

Setup: Samples are drawn uniformly per class ($K = 1$: 2,000 total; $K = 2, 5$: 1,000–500 per class; $K = 10$: 500 per class). VAE ($\beta = 4$, hidden dims [512, 256], 75 epochs) is swept over $m \in \{2, 4, 8, 12, 16, 20, 32, 64\}$.

Table 22: EY: Class-Restricted MNIST — AU_{sat} and TwoNN ID vs. class count (VAE, $\beta = 4$, 75 epochs, $m = 64$)

Configuration	K	AU_{sat}	TwoNN ID	MLE ID
1 class (digit “1”)	1	21	5.10	4.65
2 classes (“0”, “1”)	2	60	5.67	6.28
5 classes (“0”–“4”)	5	50	7.48	8.27
10 classes (all)	10	27	8.89	9.67
(Ref.) E4: 10 cl., 300 ep.	10	36	10.02	11.85

Results: TwoNN ID increases monotonically with class count ($5.10 \rightarrow 5.67 \rightarrow 7.48 \rightarrow 8.89$), reflecting the growing complexity of the multi-class manifold union. At full convergence: 1-class $\text{AU}_{\text{sat}} \approx 7$ (150-epoch preliminary run) vs. 10-class $\text{AU}_{\text{sat}} = 36$ (E4) is a ≈ 5 -fold increase, consistent with Manifold Entanglement. The 75-epoch AU_{sat} values are preliminary (non-monotone due to incomplete convergence); a full sweep is left as future work.

Conclusion: The monotone TwoNN ID increase supports the Manifold Entanglement interpretation: multi-class manifold structure genuinely increases effective dimensionality, distinguishing it from capacity leak.

6 Discussion

6.1 Answer to RQ1: Alignment of m^* with $\hat{d}_{\text{ID}}$

Main conclusion: On synthetic manifolds, m^* aligns well with $\hat{d}_{\text{ID}}$ (Swiss Roll: $m^* = 2 = d_{\text{true}}$; Torus: $m^* = 3 = d_{\text{emb}}$).

TwoNN underestimation and mitigation: The underestimation at $m = d_{\text{true}}$ (Swiss Roll $m = 2$: ID = 1.15) is a known finite-sample effect [3]; accurate estimates at $m > d_{\text{true}}$ ($m = 3$: ID = 1.94). Recommend confirming the stable region across multiple m values.

MNIST: TwoNN ID = 10.02 and MLE ID = 11.85 from AE ($m = 256$) suggest effective ID ≈ 10 –12. Gradual-type AU analysis identifies $m^{\text{knee}} = 12$, consistent with $\hat{d}_{\text{ID}}$.

6.2 Answer to RQ2: AU_{sat} and Topological Embedding Dimension

Under standardized preprocessing, Torus and Möbius Band show VAE $\text{AU}_{\text{sat}} = 3$, matching $d_{\text{emb}} = 3$ (Table 23). AU_{sat} is sensitive to input scale and should be interpreted as an *empirical correspondence under standardized preprocessing*, not a strict topological equality.

Table 23: RQ2 Summary: AU_{sat} vs. Topological Embedding Dimension

Manifold	d_{ID}	π_1	d_{emb}	AU_{sat} (std.)
Torus T^2	2	$\mathbb{Z}^2$	3	3 ✓(main)
Möbius Band	2	$\mathbb{Z}$	3	3 ✓(main)
Swiss Roll	2	$\{1\}$	2	3 × (exception)
Circle S^1	1	$\mathbb{Z}$	2	2 ($m \geq 6$) △

6.3 Answer to RQ3: Isometry and Comparison with CAE

IsometricAE achieves $\kappa = 1.055 \pm 0.005$ (lowest among 4 models), demonstrating practical achievability of isometry. CAE shows the highest $\kappa = 1.907 \pm 0.059$: encoder-side regularization is insufficient for decoder isometry. Trustworthiness measures only ordinal rank preservation; κ is the primary metric for RQ3.

6.4 Practical Design Guidelines

1. Train a reference AE with large m (e.g., $m = 256$) and apply TwoNN/MLE to obtain $\hat{d}_{\text{ID}}$.
2. Sweep $\mathcal{M}$ with VAE and observe AU pattern. For sharp saturation: adopt m^{sat} ; for gradual increase: use m^{knee} .
3. Finalize with CKA and MSE as auxiliary criteria (Eqs. (6), (7)).

For topologically nontrivial data ($\text{AU}_{\text{sat}} > d_{\text{ID}}$), consider setting m^* slightly above $\hat{d}_{\text{ID}}$.

6.5 Limitations and Future Work

1. Experiments limited to fully-connected AEs; Conv-AE needed for CIFAR-10.
2. EB could be extended to Klein Bottle, projective spaces.
3. λ_{iso} sensitivity analysis is left as future work.
4. AU threshold $\delta = 10^{-2}$ may need recalibration for different data scales.
5. TwoNN underestimation bias quantification remains open.
6. $\tau_{\text{knee}} = 0.25$ universality across diverse domains is unverified.
7. IsometricAE computation scales as $O(m)$ additional backward passes; impractical for CIFAR-10 ($\sim 64\times$) or ImageNet ($\sim 3000\times$) without approximations.
8. Multi-seed validation of MNIST E4 and E6 is future work.
9. Distinguishing Manifold Entanglement from capacity leak requires latent traversal studies.
10. Confirming Manifold Entanglement requires persistent homology analysis [7].
11. Quantifying isometry bias in the reference AE latent space for MNIST is unresolved.
12. AU_{sat} preprocessing dependence requires normalization-invariant regularization.

7 Conclusion

This paper proposed an ID estimation–based design principle for AE bottleneck dimensions, validated on synthetic manifolds and MNIST.

RQ1: $\hat{d}_{\text{ID}}$ aligns well with m^* on synthetic manifolds, functioning as a strong prior indicator (with limited accuracy on real data).

RQ2: On topologically nontrivial manifolds (Torus, Möbius Band), VAE AU_{sat} corresponds to the topological minimum embedding dimension under standardized preprocessing (with exceptions).

RQ3: IsometricAE achieves $\kappa \approx 1$, demonstrating practical achievability of the isometry prerequisite.

A secondary finding quantifies the temporal separation of global and local structure formation in MNIST training.

Practical three-step design flow: (1) Apply TwoNN/MLE to a reference AE ($m = 256$) to obtain $\hat{d}_{\text{ID}}$; (2) sweep VAE over $\mathcal{M}$ and observe AU pattern; (3) finalize m^* using CKA and MSE as auxiliary criteria.

This work contributes experimental evidence to the growing body of research on ID analysis [13, 15] and persistent homology–ID integration [14, 7].

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A Exponential Saturation Model and Sensitivity Analysis for τ_{knee}

Information-Theoretic Motivation

If $\text{AU}(m) \approx A(1 - e^{-\alpha m})$, then the growth rate $r(m) \propto e^{-\alpha m}$ decays exponentially. The natural inflection occurs around $r(m) \approx r_{\text{max}}/e \approx 0.37 \cdot r_{\text{max}}$, and $\tau_{\text{knee}} = 0.25$ is positioned as a conservative lower bound of this theoretical inflection region $[0.25, e^{-1}]$. Thus $r(m) < 0.25 \cdot r_{\text{max}}$ means “information utilization efficiency below 25% of peak,” indicating effective dimension usage has essentially ceased.

Sensitivity Analysis

For MNIST VAE, $r(12) = 0.0$ and $r(16) = 0.75$, so any $\tau_{\text{knee}} \in (0, 0.75)$ selects $m^{\text{knee}} = 12$ (robust over a wide interval of length 0.75). Across all datasets (Swiss Roll, Torus, MNIST), m^{knee} identification is unchanged for any $\tau_{\text{knee}} \in [0.1, 0.5]$.

Multi-seed experiments (EX) confirm that $r(12)$ drops sharply in all seeds, yielding $m^{\text{knee}} = 12$ consistently (representative values: $m = 12$ AU = 7.7 ± 0.6 , $r(12) = 0.00 \pm 0.00$, $r(16) = 0.8 \pm 0.2$).

Cross-Dataset Validation of τ_{knee} Robustness (Experiment EZ)

To assess universality, we evaluate Fashion-MNIST and synthetic Gaussian mixtures (5-/10-component, $d_{\text{true}} = 5$, $D = 50$) using VAE ($\beta = 4$), comparing m^{knee} across $\tau \in \{0.1, 0.2, 0.25, 0.3, 0.4, 0.5\}$.

Table 24: EZ: τ_{knee} Cross-Dataset Robustness

Dataset	AU _{sat}	TwoNN	$\tau=0.1$	$\tau=0.2$	$\tau=0.25$	$\tau=0.3$	$\tau=0.4$	$\tau=0.5$
Fashion-MNIST	10	7.00	20	20	20	8	8	8
GMix (5-comp)	1	4.27	2	2	2	2	2	2
GMix (10-comp)	3	3.02	2	2	2	2	2	2
MNIST (ref.)	36	10.02	12	12	12	12	12	12

Fashion-MNIST: AU increases gradually from 1 to 10 (gradual-type, like MNIST; TwoNN ID ≈ 7.0). The normalized rate $r(m) = 0.25$ exactly at $m = 8, 12, 16$, causing boundary sensitivity: $\tau = 0.25$ yields $m^{\text{knee}} = 20$, while $\tau = 0.30$ gives $m^{\text{knee}} = 8$. The qualitative conclusion (“ $m = 8$ –20 is the slow-growth region”) is consistent across all τ . Recommend supplementing with the second-order difference criterion when $r(m)$ values cluster near τ_{knee} .

Gaussian mixtures: The $[0, 1]$ normalization amplifies effective KL strength, causing posterior collapse (AU_{sat} = 1–3, much less than $d_{\text{true}} = 5$). $m^{\text{knee}} = 2$ is stable across all τ (trivially, due to collapse). Component-wise standardization would recover more meaningful AU_{sat} values.

Summary: $\tau_{\text{knee}} = 0.25$ is fully robust for MNIST; boundary sensitivity arises for Fashion-MNIST at $\tau = 0.25$ –0.30. Gaussian mixture results reflect preprocessing-induced collapse rather than τ sensitivity.